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Unlocking Creativity With Artificial Intelligence (AI): Field and Experimental Evidence on the Goldilocks (Curvilinear) Effect of Human–AI Collaboration

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Humans create artificial intelligence (AI), but can AI help humans create? Numerous studies show how AI enhances productivity; however, little is known about creativity—another aspect of performance that requires higher level problem-solving. To understand how AI affects the creative process, I conducted two experiments by assigning 139 business professionals and 319 working adults to collaborate in varying degrees with ChatGPT on an entrepreneurial challenge. In contrast to the well-documented positive correlation between AI usage and productivity and early studies suggesting the same for creativity, the present research shows a Goldilocks (curvilinear) effect: Moderate (vs. low or high) human–AI collaboration increases creative performance. This effect, holding across general creativity rated by human judges (either the crowdsourced public or specific trained individuals), business values by entrepreneurs, and AI-evaluated creativity, is explained by the generation of new diverse ideas (i.e., knowledge diversity) rather than problem restructuring during the brainstorming stage. I further replicate the Goldilocks phenomenon with multisource–multiwave surveys among workers in the creative industries ($N = 188$). Overall, these findings provide timely insights to the broader public regarding the effective approach to working with AI tools, such as ChatGPT, in daily and professional life. This research emphasizes the importance of striking the right balance—not too little, not too much—when working with AI technologies.

Public Significance Statement

Recent scientific discoveries have suggested that increased collaboration with artificial intelligence (AI) enhances not only productivity but, potentially, creative performance. In this research, I refine this popular notion. Through two carefully designed experiments involving business professionals and working adults engaging in an entrepreneurial challenge with ChatGPT, an AI chatbot that has grown in popularity since November 2022, I find that only the right (i.e., moderate) amount of human–AI collaboration maximizes creative outputs, whereas low and high AI usage hinder creativity. This phenomenon, which I term the Goldilocks effect, occurs because moderate (vs. low or high) human–AI collaboration enables individuals to effectively generate with AI a more diverse range of ideas and new knowledge, leading to breakthroughs during brainstorming. In a follow-up field survey study, I observe this Goldilocks phenomenon among workers in creative industries. My findings provide timely insights to the broader public regarding the effective approach to working with AI tools, such as ChatGPT, in daily and professional life.

Keywords: future of work (artificial intelligence, ChatGPT), human–artificial intelligence collaboration, creative problem-solving, brainstorming and idea generation, Goldilocks (curvilinear) effect

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The preregistrations for Study 2 (https://aspredicted.org/MN8_XZF) and Study 3 (https://aspredicted.org/DC2_WKN) are available on AsPredicted, research materials are attached in the Supplemental Material, and data and code are available on the Open Science Framework at https://bit.ly/GoldilocksEffect. An earlier version of this research was presented at the 4th Artificial Intelligence in Management Conference.

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As rapid advancements in artificial intelligence (AI) technologies, including the latest groundbreaking system developed by OpenAI, ChatGPT, continue to reshape our world (Amabile, 2019; Berg et al., 2023; Grossmann et al., 2023), a surge of research has been undertaken to showcase their advantages for workers. Existing studies from psychology, management, behavioral economics, and related fields have directed much attention to how AI increases productivity (Brynjolfsson et al., 2025; Frank et al., 2019; Noy & Zhang, 2023; Wang et al., 2024)—a crucial pillar of performance that centers on the quantitative aspect of labor effort. However, another vital aspect that necessitates higher level problem-solving and, importantly, lies at the core of our humanity remains surprisingly underexplored: creativity.

How does AI, with its capacity to generate original content (e.g., texts, visuals, arts) across various domains, affect human creative decision making? Does it support or, conversely, replace creativity? These relatively unaddressed fundamental questions have far-reaching implications for theory and societal practice (Amabile, 2019; Anantrasirichai & Bull, 2022; Berg et al., 2023; Epstein et al., 2023; Kittur et al., 2019; Rafner et al., 2023), especially considering a 2023 survey from *Nature* indicated that approximately 60% of scientists as well as the general public are now engaging in its use, such as ChatGPT, DALL·E, and Stable Diffusion, for creative purposes (Owens, 2023).

Creativity, defined psychologically as generating ideas that are both novel and useful for innovation (Amabile, 1982; Frith et al., 2021), presents a challenge when applying theoretical insights gleaned from the economic productivity effects of AI (i.e., efficiency based on volume of work and time spent; Burgmer et al., 2019; Zhang et al., 2020). Despite potential parallels suggested by preliminary findings that AI holds the potential to augment creativity (Boussioux et al., 2024; Doshi & Hauser, 2024; Hitsuwari et al., 2023; Jia et al., 2024; Lee & Chung, 2024), I contend that the expected paradigm of how AI use influences creative endeavors may differ from a conventional positive effect (well-documented in productive performance and recently observed in creative performance as well). Given the intricate dynamic of creative decision making (Baer & Oldham, 2006; Shin & Grant, 2021), in this research, I depart from prior literature by introducing a new perspective that aligns with recent psychological theory about "too much of a good thing" (Grant & Schwartz, 2011). Specifically, I propose that "just the right (moderate) amount" of the usage of AI maximizes creative outcomes, whereas both low and high levels of collaboration with AI can be suboptimal and potentially detrimental.

Minimal involvement with AI may lead individuals to solely focus on their own viewpoint, limiting exposure to different perspectives (e.g., from generative AI). Since high-quality ideas often emerge from diverse perspectives and significant deviations, this restricted cognitive flexibility to explore alternative angles as well as different information processing can hinder the generation of innovative ideas (Kittur et al., 2019; Zhu et al., 2023). Additionally, the brainstorming process usually takes time (Shin & Grant, 2021). Individuals who have less collaboration with AI must allocate substantial time to generate a diverse set of ideas themselves, leaving insufficient time for selecting and refining the best ideas. Such time pressure and cognitive challenges (e.g., consumed cognitive resources) can also make one feel less competent in creativity. Consequently, these limitations impede the activation of new, diverse knowledge necessary for breakthroughs (Tierney & Farmer, 2002).

In contrast, during moderate human–AI collaboration, individuals can effectively utilize AI as a tool while still leveraging their own human intelligence. Doing so offers better problem analysis and time allocation for generating ideas. For example, contemporary AI algorithms such as ChatGPT, functioning like human agents capable of analyzing problems logically and producing various ideas/suggestions (Binz & Schulz, 2023), may serve as sources of inspiration for human creators. This facilitates the generation of unconventional concepts and further streamlines the brainstorming process (Kittur et al., 2019). This approach can not only save time but also enhance the sense of competence in problem-solving. Collectively, these factors contribute to a collaborative synergy of AI-augmented creative output.

This beneficial effect on creativity, nonetheless, is unlikely to extend to a high degree of collaboration. First, although high AI usage might facilitate idea generation in the beginning, AI models, learning from historical big data, often exhibit a standardized response pattern (Park et al., 2024). If initial ideas are predominantly conceptually alike—being incremental and lacking knowledge divergence—they are likely to impair final creative outputs (Audia & Goncalo, 2007). Moreover, heavy reliance on AI tools, while overlooking workers' own inventive potential and perspective to dissect problems, may fixate workers on AI's analytical approaches and further produce an anchoring effect (anchored on AI's expected and less distinctive ideas), which can adversely affect creative outcomes (Chen & Chan, 2024; see also Reis et al., 2024). As a result, humans may experience a diminished sense of competence. They can also perceive AI as a potential threat capable of replacing their skills (Jia et al., 2024) or limiting opportunities for creative expression (Epstein et al., 2023). In such cases, the incorporation of AI into organizational tasks may not augment, but rather undermine and replace, human creativity.

Building on these rationales, I predict the existence of a Goldilocks effect: a *curvilinear* (nonmonotonic inverted-U-shaped) relationship[1] between human–AI collaboration and human creativity. To causally investigate this Goldilocks phenomenon and explore its underlying mechanisms (e.g., cognitive processes) as well as other associated psychological benefits that support my theory and hypothesis (e.g., sense of competence), the current approach departs from existing studies on human–AI collaboration. Prior research focuses experimentally on providing humans access to AI or not (Boussioux et al., 2024; Brynjolfsson et al., 2025; Fügener et al., 2022; Hitsuwari et al., 2023; Jia et al., 2024; Lee & Chung, 2024; Noy & Zhang, 2023; Wang et al., 2024), which implicitly assumes and could only find "all-or-none" for positive relationships. In this article, I overcome this shortcoming by directly manipulating varying degrees of AI usage in a lab-in-the-field experiment to provide, first, causal evidence for examining possible curvilinear effects of AI on creative performance. Further, I replicate the results of the Goldilocks effect through a preregistered online experiment among working adults with improved measurements (direct replication) and a preregistered field study among professionals working in the creative industries

[1] For some readers, curvilinear effects are technically interaction effects in statistics, as "a quadratic relationship in X is the same as a relationship between X and Y moderated by X—the nature of the association depending on the value of X" (Dawson, 2014, p. 15). However, in this research, I treat curvilinear effects as conceptually distinct from traditional interaction effects, as they do not reflect moderation by a separate variable (e.g., Z) typically seen in studies testing interactions.

(conceptual replication) to generalize my findings in the real-world labor market.

Overall, the present research demonstrates experimental (Study 1 and Study 2) and field evidence (Study 3) that moderate (as opposed to low or high) collaboration with AI results in the highest creativity. In doing so, I contribute to the literature by challenging previous assumptions about the potential (linear) relationship between AI and creativity and offering a novel perspective as well as a methodological approach to account for the curvilinear relationship. Moreover, this work unveils key creative processes during human–AI collaboration, including knowledge diversity (instead of problem restructuring), as I explain below, that drive this Goldilocks effect.

Transparency and Openness

Research materials (including full measures) are available in the Supplemental Material, and data and code are available on the Open Science Framework at https://bit.ly/GoldilocksEffect. Studies 2 and 3 were preregistered. Any deviations from the preregistered plans are described within each study and summarized in the Supplemental Material.

Study 1: Laboratory-in-the-Field Experiment

Method

I conducted a randomized laboratory-in-the-field experiment at an international university business school in Canada in an effort to assess the initial effects and inform the follow-up studies. This lab-in-the-field approach combined a validated, standardized paradigm from the lab with business adults who held relevant backgrounds in the field (e.g., bachelor of commerce, master of business administration, business alumni) for both internal and external validity purposes (Gneezy & Imas, 2017). Power analysis indicated that 111 participants were required to detect a medium effect size with .80 power across three conditions (Faul et al., 2007). A total of 148 participants signed up for an online experimental session. After excluding those who failed an attention check or did not submit their creative proposal, the final sample was 139 business professionals (61 reported as female, 72 as male, six preferred not to say or did not report; M_{age} = 25.19, SD_{age} = 3.20), and they had an average of 5.65 years (SD = 3.82) of work and/or managerial experience.

Participants were randomly assigned to one of three experimental conditions simulating real collaboration with AI on tasks in work-related settings: low human–AI collaboration versus moderate human–AI collaboration versus high human–AI collaboration. They worked with ChatGPT-3.5, an assistive AI chatbot, at varying levels of collaboration to complete a two-phased creative task, which was a problem-solving challenge utilized in an entrepreneurship class at Stanford University (Shin & Grant, 2021). While the entire study was a 30-min online session (including task introduction, a 15-min creative task, and a posttask survey containing manipulation check, exploratory measures, and control variables), overall, participants were highly engaged in the experiment and enjoyed the task, as evidenced by a mean of 5.26 out of 7 for task interest.

Treatment

The entrepreneurial challenge in the experiment was to assist a student named Mike in generating original ideas and novel strategies for a start-up business. The objective of the assignment was to maximize earnings within a 3-hr period of time with a seed fund of $10, which Mike could allocate toward purchasing materials or resources. For instance, Mike could use the seed money to buy cups and lemons and sell lemonade. Participants' goal in this task was to brainstorm and help Mike develop and execute a winning idea so that he could present his business results to the class. This task contained two phases. In the brainstorming phase (Phase 1), participants were instructed to write down all ideas that came to mind in the designated space. This part allowed us to assess cognitive processes related to the underlying mechanisms of creativity, including problem restructuring and knowledge diversity during brainstorming (Shin & Grant, 2021; Zhu et al., 2023). Following the brainstorming phase (Phase 2), participants needed to select their most creative idea and compose an email to Mike explaining the chosen idea in detail, which enabled us to evaluate the creativity of the final proposal. Participants had a total of 15 min to complete this task. They had the freedom to distribute their time within the timeframe across brainstorming and writing the email outlining their most creative proposal to Mike.

Moreover, to incorporate the AI component into this experiment, all participants had access to and collaborated with ChatGPT-3.5 in this task, which I positioned as "an AI that can engage in natural language conversations with users and respond to a wide range of topics, from general knowledge to specific queries, to assist you [i.e., them]." In line with Lee and Chung (2024), they could ask ChatGPT-3.5 to help generate creative ideas (or polish existing ones). To manipulate different levels of human–AI collaboration, I randomly assigned three treatments to participants in terms of their degree of access to ChatGPT. In the low human–AI collaboration treatment, I advised participants to limit their usage of ChatGPT to one time (each question or request counted as one time). In the moderate human–AI collaboration treatment, participants could use ChatGPT between four and six times. In the high human–AI collaboration treatment, participants were told to interact with ChatGPT at least nine times. These treatment groups were determined based upon a pilot (N = 69), where people completed the same task and indicated that the ideal number of collaborations with ChatGPT for this creative task was 4.04 times (SD = 2.40), whereas 0.75 times (SD = 1.01) was deemed too few and 7.42 times (SD = 5.11) was too many.

In an effort to prevent confounds in my experimental design, I provided all participants with access to ChatGPT (those in the low human–AI collaboration group also had access rather than no access). As such, this assured that the findings were caused solely by the varying degree of collaboration with AI, instead of the difference regarding whether one could access ChatGPT or not. Further, I ensured that all recruited participants had at least some experience and knew how to work with ChatGPT (M = 8.37 weeks of experience, SD = 7.06). Thus, none of them were completely new to this AI chatbot. Together, these measures were taken in this behavioral experiment (see also robustness checks below) not merely to furnish robust causal evidence but also to closely reflect the real-world phenomenon of cooperating with AI technologies on problem-solving and decision making.

Measurement of Creativity

Since accurate measurement of creative outcomes is critical for testing the proposed curvilinear relationship, Study 1 operationalized

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the creativity of the final proposal (Phase 2) in three ways for robustness: human-rated creativity, entrepreneurial business value, and AI-rated creativity.

Human-Rated Creativity. I recruited two independent judges who were blind to the research purpose and experimental conditions to evaluate participants' creative proposal to Mike. Each judge received proper training, which emphasized that creative ideas should be both novel and useful (Grant & Berry, 2011), and rated the email proposals on a 7-point scale (1 = *not at all creative*, 7 = *extremely creative*). Given two judges achieved good interrater reliability (intraclass correlation coefficient, ICC = .73), I averaged the two ratings for an overall score. An idea about setting up a lemonade stand was rated as not creative, while ideas about organizing a mini-treasure hunt event with entrance fees or providing a customized memorable performance to customers for celebrating special occasions (which the participant called "Singing and Dancing Gram") were deemed very creative.

Entrepreneurial Business Value. I assembled a panel of two professional judges, distinct from those rating general creativity, to assess the business aspects of creative ideas (Girotra et al., 2010). The panel judges were professionals with relevant work experience, such as entrepreneurship or investment. I provided detailed instructions by having them independently evaluate the final proposals from an entrepreneurship perspective (e.g., considering those ideas as potential, innovative start-up businesses) and assign a score for their business value (1 = *lowest business value*, 10 = *highest business value*; Girotra et al., 2010). Once again, I calculated the interrater reliability (ICC = .71) and averaged the two ratings to obtain a composite.

AI-Rated Creativity. In the last measure of creativity, I drew on emerging insights on large language models (LLMs) in psychological interpretation (Demszky et al., 2023; Rathje et al., 2024) and ventured into exploring the potential of machine intelligence for subjective ratings such as creativity. I provided GPT with the same instructions used to train human judges in evaluating creative ideas. I tasked GPT with assessing the creativity criteria of all participants' creative proposals based on both novelty and utility through a 7-point scale (1 = *not at all creative*, 7 = *extremely creative*). To the best of my knowledge, as relatively few LLM studies have focused on rating creativity (as opposed to other less subjective constructs), this measurement serves as a conservative—but exciting—test of theory.[2]

Psychological Mechanisms (Cognitive Processes)

To investigate the mechanisms underlying the proposed curvilinear effect of human–AI collaboration on creativity, I hired two separate coders who were blind to the research to capture the cognitive processes during the brainstorming stage (Phase 1). The two fundamental thought processes the present research focused on are problem restructuring and knowledge diversity (Shin & Grant, 2021; Zhu et al., 2023).

Problem Restructuring. The thought process of problem restructuring is crucial for fostering creative outcomes because reframing the problem from a different perspective allows participants to think outside the box and become more creative (Shin & Grant, 2021). In the original case of the Stanford classroom, creative students made reservations at popular restaurants on weekend nights to earn hundreds of dollars by selling those spots, without being constrained by $10 seed money in finding innovative solutions. Therefore, Study 1 measured problem restructuring by asking two coders to tally how many generated ideas from each participant went beyond the initial framing of the problem and relaxed its constraints (i.e., focusing on maximizing earnings, rather than considering the use of $10 seed fund to generate money). Two coders also counted the total number of ideas generated during brainstorming. I thus calculated the percentage of brainstormed ideas that restructured the problem. The interrater reliability was high (ICC = .89), and I averaged the two codings.

Knowledge Diversity. Knowledge diversity is another critical cognitive process that emerges during brainstorming sessions, characterized by the generation of distinct categories, perspectives, concepts, or approaches in ideas. It stems from divergent thinking (or cognitive flexibility), which involves breaking away from conventional or linear thought patterns and exploring various potential perspectives, thereby enhancing creativity (Zhu et al., 2023). This process stimulates the activation of new knowledge, facilitating the discovery of novel information and the creation of a diverse set of original and feasible ideas. This is pivotal for the final selection and further development of the best creative idea, as it provides a broader range of options to choose from. To assess the diversity of knowledge, Study 1 engaged two coders who evaluated the level of diversity exhibited by the brainstormed ideas (1 = *not at all diverse*, 7 = *extremely diverse*). A rating of 1 suggests that all ideas from a participant were derived from or based upon the same knowledge, such as proposing two similar ideas: selling lemonade and merchandising baked goods. On the other hand, a rating of 7 signifies that the ideas encompassed a broad variety of new knowledge. For instance, a participant suggested two diverse ideas: gambling/buying lottery and offering services like house cleaning. The two scores from coders were averaged (ICC = .79).

Experimental Checks

Manipulation Check. As a validation of the human–AI collaboration manipulation, Study 1 recorded the number of times participants cooperated with ChatGPT based on their honest response to the question, "How many times did you use ChatGPT during this task?" (Each question or request counts as one time). The manipulation was effective, $F(2, 136) = 107.79$, $p < .001$, $\eta^2 = .61$, such that participants in the low human–AI collaboration treatment group utilized the fewest times ($M = 1.50$, $SD = 0.89$), followed by the moderate ($M = 4.20$, $SD = 1.60$) and high human–AI collaboration ($M = 7.85$, $SD = 3.11$), with all differences significant at $p < .001$.

Robustness Checks. For potential confounds that might influence participants' creative experience in the task, participants reported their familiarity with ChatGPT (1 = *not at all familiar*, 7 = *extremely familiar*), how long they had been using ChatGPT (in weeks), and task interest regarding whether they enjoyed the task and found it fun (Grant & Berry, 2011; 1 = *strongly disagree*, 7 = *strongly agree*; $\alpha = .97$). As the results held with and without these variables, passing

[2] Overall, the three distinct ratings of creativity show some positive correlations with one another ($r_{\text{human-rated creativity, AI-rated creativity}} = .23$, $p = .006$; $r_{\text{human-rated creativity, business value}} = .13$, $p = .13$; $r_{\text{business value, AI-rated creativity}} = .10$, $p = .24$).

robustness tests (see Supplemental Material), I only report results without controls below.[3]

Results

Table 1 displays the means and standard deviations of Study 1 variables by condition. A full list of the best ideas generated for final proposals is provided in Table 2.

Creativity Effects

A one-way analysis of variance (ANOVA) showed a significant effect of human–AI collaboration on human-rated creativity of the final proposals, $F(2, 136) = 3.41$, $p = .036$, $\eta^2 = .05$. Planned comparisons indicated a curvilinear effect, with the proposals in the moderate human–AI collaboration being rated as more creative by human judges ($M = 4.02$, $SD = 1.21$) compared to those in the low human–AI collaboration ($M = 3.49$, $SD = 1.25$), $t(136) = 2.05$, $p = .042$, Cohen's $d = .43$, and high human–AI collaboration conditions ($M = 3.39$, $SD = 1.28$), $t(136) = 2.43$, $p = .016$, $d = .50$. No significant difference between low and high human–AI collaboration was observed (Figure 1).

Examining the business value of the final proposals, an ANOVA demonstrated a significant effect of human–AI collaboration, $F(2, 136) = 4.10$, $p = .019$, $\eta^2 = .06$. Planned comparisons revealed that the proposals written by participants in the moderate human–AI collaboration were considered to have higher business values by professional judges ($M = 5.67$, $SD = 2.04$) than those in the low ($M = 4.60$, $SD = 1.89$), $t(136) = 2.58$, $p = .011$, $d = .54$, and high human–AI collaboration treatments ($M = 4.69$, $SD = 2.07$), $t(136) = 2.37$, $p = .019$, $d = .49$. I did not discover a significant difference in business value between the low and high human–AI collaboration treatments. Figure 2 presents the results.

Last, I was interested in the effect of human–AI collaboration on AI-rated creativity. An ANOVA showed a nonsignificant effect, $F(2, 136) = 2.85$, $p = .061$, $\eta^2 = .04$. Despite the lack of conventional significance, to explore the potential direction, I proceeded with planned comparisons and found that participants in the moderate human–AI collaboration group wrote significantly more creative ideas as assessed by the ChatGPT rater ($M = 3.91$, $SD = 1.26$) than the low human–AI collaboration group ($M = 3.33$, $SD = 1.17$), $t(136) = 2.38$, $p = .019$, $d = .50$, but not the high human–AI collaboration group ($M = 3.66$, $SD = 1.11$), $t(136) = 1.03$, $p = .30$, $d = .21$. Additionally, as shown in Figure 3, no difference in AI-rated creativity was detected between the low and high human–AI collaboration groups.

Mediating Roles of Problem Restructuring and Knowledge Diversity (Cognitive Processes)

There were no significant differences in problem restructuring during brainstorming between treatments, $F(2, 136) = 0.10$, $p = .90$, $\eta^2 = .001$.[4] Conversely, I found a significant effect of treatments on knowledge diversity in brainstormed ideas, $F(2, 136) = 5.87$, $p = .004$, $\eta^2 = .08$, and observed a curvilinear relationship: Participants' overall ideas in the brainstorming stage contained significantly more diverse knowledge in the moderate human–AI collaboration treatment condition ($M = 5.10$, $SD = 1.00$) than in both low ($M = 4.30$, $SD = 1.30$), $t(136) = 3.37$, $p < .001$, $d = .70$, and high human–AI collaboration ($M = 4.57$, $SD = 1.07$), $t(136) = 2.23$, $p = .027$, $d = .46$.

I further conducted mediation analyses for curvilinear effects (Hayes & Preacher, 2010). The instantaneous indirect effect of human–AI collaboration on (human-rated) creativity through problem restructuring was not significant for low ($\theta = .001$, 95% confidence interval, CI [−.07, .11]), moderate ($\theta = .001$, 95% CI [−.02, .03]), and high human–AI collaboration ($\theta = .0001$, 95% CI [−.09, .09]), suggesting that problem restructuring did not account for the curvilinear creativity effect. In contrast, the instantaneous indirect effect through knowledge diversity was positive at low human–AI collaboration ($\theta = .38$, 95% CI [.11, .79]), nonsignificant at moderate human–AI collaboration ($\theta = .04$, 95% CI [−.02, .13]), and negative at high human–AI collaboration ($\theta = -.31$, 95% CI [−.68, −.08]). These findings mean that from a low to moderate degree of collaboration with AI, more human–AI collaboration increased creativity via facilitating knowledge diversity, yet from a moderate to high degree of collaboration with AI, more human–AI collaboration decreased creativity via hindering knowledge diversity. Taken together, knowledge diversity during brainstorming explained the curvilinear relationship.

Supplemental Evidence

Study 1 also uncovered further evidence that supports my theory of the observed curvilinear phenomenon for creative processes, such as sense of competence and collective psychological ownership. Please refer to the Supplemental Material (and Supplemental Table S1) for these additional results.

Study 2: Replication Experiment

Method

Though Study 1 as a preliminary test provided initial support for my hypothesis across various measures of creativity, I wanted confidence to ensure that these results were replicable. Accordingly, in line with Open Science practices, I conducted a preregistered direct replication of Study 1 (https://aspredicted.org/MN8_XZF), focusing on the Goldilocks effect of human–AI collaboration on (general) creativity and its central underlying mechanisms. The design of Study 2 was the same as in Study 1, featuring a 15-min two-phased entrepreneurial problem-solving task with three between-subjects conditions: low versus moderate versus high human–AI collaboration. To be conservative, Study 2 aimed to more than double the sample size from Study 1 and recruited 338 working adults in the United States via CloudResearch. After excluding those who failed an attention check or did not brainstorm or submit a creative proposal,

[3] Another potential confounding factor is *how* participants use AI, independent of the amount of usage. Past studies have shown that individuals in creative tasks can use LLMs as ghostwriters to "generate an initial set of content to kickstart the creative process" and help complete the writing task or as sounding boards to "evaluate and critique users' creative work" and provide feedback for improvement (Chen & Chan, 2024, p. 9102). I did not find significant differences in how participants leveraged ChatGPT, as most of them treated it more as a ghostwriter modality to help generate creative ideas and assist with writing the final creative proposal, suggesting that the reported findings are largely determined by the degree of human–AI collaboration relative to the way AI is used.

[4] I also did not find an effect of human–AI collaboration on the (total) number of ideas generated during the brainstorming process ($M = 8.10$, minimum = 1, maximum = 31).

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Table 1
Means and Standard Deviations by Experimental Condition in Study 1 and Study 2

Creative task Experimental treatment: Human–AI collaboration	Phase 2: Business proposal Creative outcome			Phase 1: Brainstorming idea Cognitive (mediating) process	
	Human-rated creativity	Business value	AI-rated creativity	Problem restructuring	Knowledge diversity
Study 1					
Low	3.49 (1.25)	4.60 (1.89)	3.33 (1.17)	.41 (.27)	4.30 (1.30)
Moderate	4.02 (1.21)	5.67 (2.04)	3.91 (1.26)	.43 (.29)	5.10 (1.00)
High	3.39 (1.28)	4.69 (2.07)	3.66 (1.11)	.44 (.29)	4.57 (1.07)
Study 2					
Low	4.04 (0.70)		4.39 (1.09)	.43 (.18)	4.45 (0.77)
Moderate	4.33 (0.73)		4.72 (1.16)	.43 (.16)	4.81 (0.68)
High	4.11 (0.77)		4.31 (1.22)	.44 (.16)	4.60 (0.84)

Note. Standard deviations are in parentheses. Study 1 gathered creativity from two raters and problem restructuring and knowledge diversity from two coders, whereas Study 2 relied on a crowdsourcing approach. AI = artificial intelligence.

the final sample in Study 2 comprised 319 participants (124 reported as female, 193 as male, two as nonbinary; M_{age} = 34.01, SD_{age} = 7.03) with an average of 12.93 years (SD = 7.77) of work experience and 31.03 weeks (SD = 19.70) of ChatGPT experience. Participants reported a high level of engagement in this online behavioral experiment ($M_{task\ interest}$ = 5.34).

Measures

In the preregistered plan, Study 2 hired two independent judges trained for rating creativity and two additional coders for problem restructuring and knowledge diversity, as in Study 1. While the results were significant (see the Supplemental Material), I decided to proceed with an improved, more robust assessment approach to gather the key variables. Leveraging the consensual assessment technique (Amabile, 1982), I crowdsourced ratings from large public samples to harness the wisdom of the crowd. Below, I describe the crowdsourcing approach in detail and present correlational evidence demonstrating its relevance to my original approach.

Creativity (Crowdsourced). A sample of 161 adults (77 reported as female, 81 as male, two as nonbinary, one preferred not to say; M_{age} = 38.28, SD_{age} = 13.56) from Prolific with some expertise in creativity (M = 6.06 years of experience in creative jobs, SD = 7.96) served as independent judges. Judges were provided with instructions that included the definition of creativity (involving both novelty and usefulness) to rate on a 7-point scale as in Study 1 (1 = *not at all creative*, 7 = *extremely creative*). Each judge rated the creativity of 20 randomly selected final proposals, and each proposal was hence assessed by an average of 10.09 different judges (SD = 0.82) to form a composite score. Overall, the crowdsourced creativity rating is significantly associated with the original rating, which relied on only two judges who assessed all email proposals (r = .18, p = .001).

AI-Rated Creativity. Given some promising indications observed in Study 1, Study 2 continued to gauge AI-rated creativity by feeding GPT and included it as an exploratory outcome. This rating correlates with the crowdsourced creativity (r = .31, p < .001), suggesting potential consistency between the algorithm and the wisdom of the general public.

Problem Restructuring (Crowdsourced). A separate Prolific sample of 163 independent coders (79 reported as female, 84 as male; M_{age} = 38.39, SD_{age} = 13.34; M = 5.85 years of experience in creative jobs, SD = 7.45) was recruited to calculate the percentage of brainstormed ideas that restructured the problem. Each coder received proper training and an example case to practice before evaluating 20 sets of ideas in the brainstorming phase. Each set of brainstormed ideas received assessments from 10.22 different coders (SD = 1.49) to form an average. Both the crowdsourcing and original approaches are highly correlated (r = .51, p < .001).

Table 2
Categories of Final Creative Proposals by Participants in Study 1

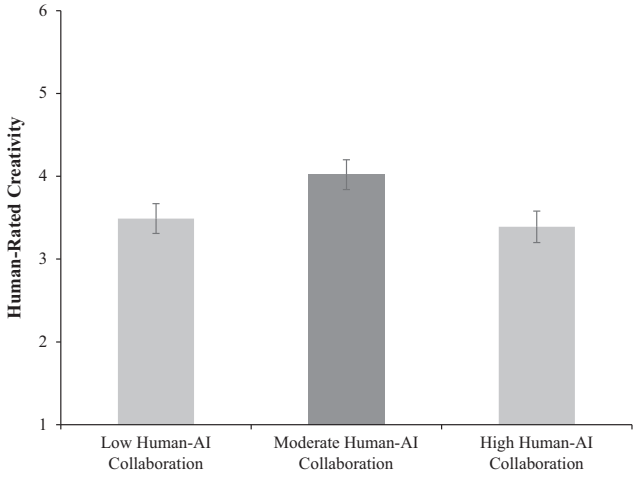
Idea category	Final proposal example	Count	%
Service and workshop	Cleaning services (Original ID No. 23) Yoga workshop (Original ID No. 1)	40	28.8
Food	Lemonade stand (Original ID No. 92)	35	25.2
General sale and resale	Dropshipping business (Original ID No. 96) Yard sale of unused items (Original ID No. 43)	29	20.9
Artisanal and crafted creation	Homemade crafts (Original ID No. 120)	16	11.5
Gambling	Online poker tournament (Original ID No. 87) Investing in stocks (Original ID No. 60)	8	5.8
Donation and busking	Fundraising event (Original ID No. 32) Playing an instrument on the street (Original ID No. 72)	7	5.0
Other	Trading items (Original ID No. 125)	4	2.9
	Total	139	100

Note. The final proposals were categorized into seven primary types of ideas by a trained research assistant who was unaware of the research purpose and generated/identified the themes on his/her own.

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Figure 1

The Effect of Human–AI Collaboration on Human-Rated Creativity in Study 1



Note. Error bars indicate the standard errors. AI = artificial intelligence.

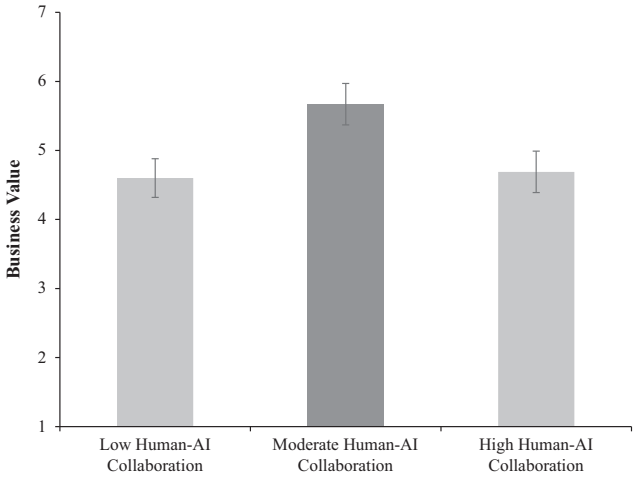
Knowledge Diversity (Crowdsourced). These 163 coders also evaluated the level of diversity exhibited by the brainstormed ideas (1 = *not at all diverse*, 7 = *extremely diverse*), independent of their assessment of problem restructuring. This crowdsourced measure relates to the original measure of knowledge diversity to a great extent ($r = .42$, $p < .001$).

Experimental Checks

Manipulation Check. Participants reported the number of times they worked with ChatGPT in this online study. The manipulation was successful, $F(2, 316) = 180.99$, $p < .001$, $\eta^2 = .53$, such that participants in the low human–AI collaboration treatment utilized it the fewest times ($M = 1.83$, $SD = 2.58$), followed by the moderate ($M = 4.50$, $SD = 2.01$) and high human–AI collaboration ($M = 8.88$, $SD = 3.40$), with all differences significant at $p < .001$.

Figure 2

The Effect of Human–AI Collaboration on Business Value in Study 1



Note. Error bars indicate the standard errors. AI = artificial intelligence.

Robustness Checks. For potential confounds, participants answered their familiarity (1 = *not at all familiar*, 7 = *extremely familiar*), experience with ChatGPT (in weeks), task interest as in Study 1 (1 = *strongly disagree*, 7 = *strongly agree*; $\alpha = .97$), trust in ChatGPT (1 = *not at all*, 7 = *extremely*), and the version of ChatGPT they used (93.1% used the GPT-3.5 free version and 6.9% the GPT-4 paid version). Results passed robustness tests (see the Supplemental Material), and I report findings without controls below.

Results

Table 1 presents the means and standard deviations of variables of interest by treatment.

Creativity

One-way ANOVA results indicated a significant effect of human–AI collaboration on crowdsourced creativity, $F(2, 316) = 4.57$, $p = .011$, $\eta^2 = .03$. As shown in Figure 4, planned comparisons revealed a curvilinear effect: Proposals in the moderate human–AI collaboration were more creative ($M = 4.33$, $SD = 0.73$) than those in the low ($M = 4.04$, $SD = 0.70$), $t(316) = 2.87$, $p = .004$, $d = .39$, and high human–AI collaboration conditions ($M = 4.11$, $SD = 0.77$), $t(316) = 2.25$, $p = .025$, $d = .31$.

I also observed in ANOVA a main effect of human–AI collaboration on AI-rated creativity, $F(2, 316) = 3.69$, $p = .026$, $\eta^2 = .02$. Specifically, participants in the moderate human–AI collaboration treatment generated significantly more creative proposals as assessed by the AI rater ($M = 4.72$, $SD = 1.16$) compared to both the low ($M = 4.39$, $SD = 1.09$), $t(316) = 2.07$, $p = .039$, $d = .28$, and high human–AI collaboration treatments ($M = 4.31$, $SD = 1.22$), $t(316) = 2.55$, $p = .011$, $d = .35$ (Figure 5).

Mediation of Problem Restructuring and Knowledge Diversity

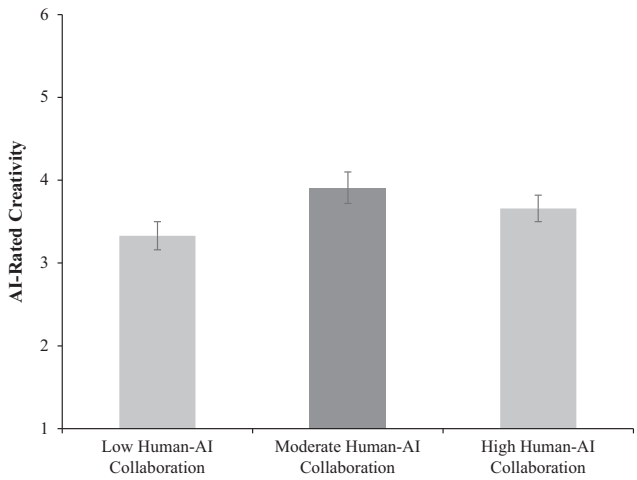
No significant differences in problem restructuring during brainstorming between treatments were detected, $F(2, 316) = 0.29$, $p = .75$, $\eta^2 = .002$. In contrast, I observed a significant (curvilinear) effect of treatments on knowledge diversity in brainstormed ideas, $F(2, 316) = 6.07$, $p = .003$, $\eta^2 = .04$, such that participants' brainstormed ideas were more knowledge-diverse in the moderate human–AI collaboration treatment ($M = 4.81$, $SD = 0.68$) as opposed to in low ($M = 4.45$, $SD = 0.77$), $t(316) = 3.47$, $p < .001$, $d = .47$, and high human–AI collaboration ($M = 4.60$, $SD = 0.84$), $t(316) = 2.04$, $p = .043$, $d = .28$.

To test for curvilinear mediation effects, I computed mediation analyses (Hayes & Preacher, 2010). The instantaneous indirect effect of human–AI collaboration on creativity through problem restructuring was nonsignificant for low ($\theta = .006$, 95% CI [−.03, .06]), moderate ($\theta = -.004$, 95% CI [−.02, .01]), and high human–AI collaboration ($\theta = -.01$, 95% CI [−.08, .02]), showing that problem restructuring did not mediate the curvilinear creativity relationship. Conversely, the instantaneous indirect effect through knowledge diversity was positive at low human–AI collaboration ($\theta = .16$, 95% CI [.07, .31]), not significant at moderate human–AI collaboration ($\theta = .02$, 95% CI [−.01, .05]), and negative at high human–AI collaboration ($\theta = -.12$,

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Figure 3

The Effect of Human–AI Collaboration on AI-Rated Creativity in Study 1



Note. Error bars indicate the standard errors. AI = artificial intelligence.

95% CI [−.27, −.04]). These findings suggest that from a low to moderate degree of human–AI collaboration, more collaboration with AI increased creativity via facilitating knowledge diversity, yet from a moderate to high degree of collaboration with AI, more human–AI collaboration decreased creativity via hindering knowledge diversity. Similar to Study 1 results, knowledge diversity during brainstorming accounted for the AI Goldilocks effect.

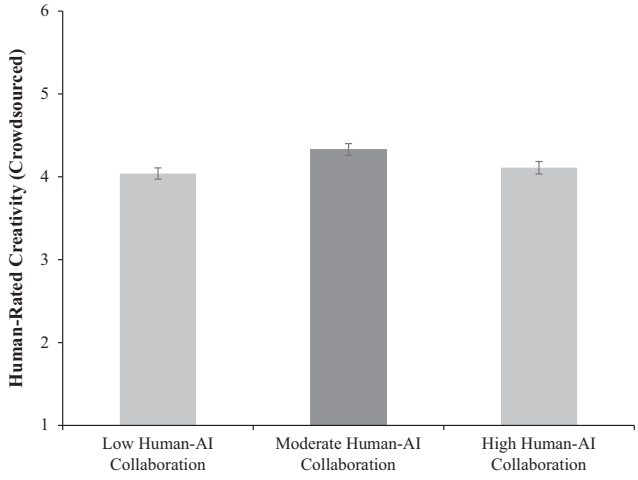
Study 3: Field Study

Method

One concern in the experimental design is that asking participants to purposefully collaborate a lot or little with AI technology creates demand effects. In an effort to conceptually replicate and generalize my findings, I conducted a preregistered field study (https://aspredicted.org/DC2_WKN) using a different field sample—creative workers in relevant industries—beyond general business professionals in Study 1 and working adults in Study 2. Before data collection, through personal networks, I interviewed creative professionals, including technical and visual artists as well as (game) designers, in an international Taiwanese game/animation company. The conversations suggested that current workers in the global creative industry are increasingly incorporating AI (in particular, generative AI like Stable Diffusion in arts due to the flexibility to adjust diffusion model parameters) into their jobs, with a wide range of AI usage patterns. Moreover, they possess the creative expertise and professional experience to approach creative tasks compared to general labor, offering nuanced insights into how AI impacts the creative processes at work (Anantrasirichai & Bull, 2022; Epstein et al., 2023). Therefore, while instances of revolutionary creativity and breakthroughs ("Big-C") might be relatively few, employees in the creative industry constitute an even more appropriate field setting to examine not only everyday creativity ("little-C") but also professional creativity ("Pro-C"), which might lead to Big-C in human society (Kaufman & Beghetto, 2009).

Figure 4

The Effect of Human–AI Collaboration on Human-Rated Creativity (Crowdsourced) in Study 2



Note. Error bars indicate the standard errors. AI = artificial intelligence.

Since my primary interest was in modeling the quadratic term as the focal predictor in multiple linear regression (rather than testing mediation due to the methodological/survey approach employed in this study), power analysis for regression indicated that 189 participants were required to detect the average effect sizes observed in Studies 1 and 2 (Cohen's $f^2 = 0.042$)[5] with .80 power (Faul et al., 2007). Aiming to reach sufficient statistical power and better external validity, I partnered with the Connect platform, which specializes in compiling nationally representative panels in the creative industry. Study 3 administered a multiwave, multisource field survey among full-time workers (and their supervisors; $N = 188$ pairs) in the creative industries in the United States.[6] Their jobs included but were not limited to (game or fashion) designers, technical and visual artists/illustrators, influencers, authors/writers, actors/performance artists, animators, technologists, and marketers. Overall, creative professionals (70 reported as female, 116 as male, two as nonbinary; $M_{age} = 37.48$, $SD_{age} = 8.15$) had 16.36 years ($SD = 9.03$) of work experience with a mean tenure of 6.68 years ($SD = 5.92$) in their current organization. They reported that creativity was an integral component of their job (a mean value of 4.69 out of 7). I measured their degree of collaboration with AI at work as well as control variables at Time 1, and approximately 1 week later (i.e., Time 2), their supervisor provided ratings of their creative work performance. This (two-wave, two-source) approach minimizes common method bias in correlational survey methodology within organizational contexts (Podsakoff et al., 2024).

Measures

Human–AI Collaboration (Time 1). At Time 1, Study 3 assessed creative professionals' degree of collaboration with AI at

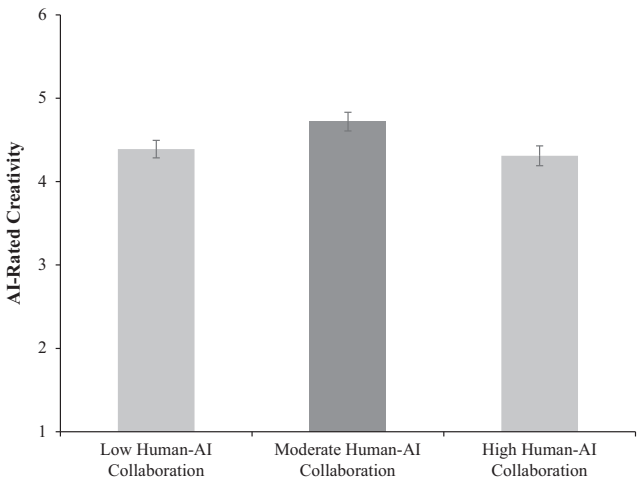
[5] At the time of the planned power analysis, the effect size extracted from Study 2 was based on my initial method of assessing creativity (ratings from two judges instead of crowdsourcing, though the two effects were very similar).

[6] While Study 3 attempted to recruit a larger sample size to satisfy power requirements, I acknowledge that this study fell slightly short of the target due to practical considerations (lower return/completion rate of surveys from field participants than originally estimated).

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Figure 5

The Effect of Human–AI Collaboration on AI-Rated Creativity in Study 2



Note. Error bars indicate the standard errors. AI = artificial intelligence.

work with three items (e.g., "I use artificial intelligence to carry out most of my job functions," "I spend most of my time working with artificial intelligence"; Tang et al., 2022) on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*; α = .94). I found empirical evidence that the experimental manipulation of the amount of AI usage (coding for low condition = −1, moderate = 0, high = 1) is positively correlated with this measure to a moderate extent ($r = .50$, $p < .001$), $F(2, 316) = 69.86$, $p < .001$, as Study 2 significant manipulation check, suggesting potential alignment between different methods.

Creativity (Time 2). To measure creativity, participants' supervisors rated them at Time 2 (i.e., 1 week later) using a nine-item list (e.g., "Generates novel, but operable work-related ideas"; Tierney et al., 1999) on a 6-point scale (1 = *not at all*, 6 = *very often*; α = .90).

Control Variables (Time 1). Due to the correlational nature of the study design, Study 3 collected several control variables in light of theoretical considerations. For primary control variables, participants reported at Time 1 their intrinsic motivation for the job (e.g., "I do this job because I find the work engaging"; Menges et al., 2017), creative requirement (i.e., "My job involves solving problems that have no obvious correct answer"; Morgeson & Humphrey, 2006), conscientiousness (e.g., "I see myself as dependable, self-disciplined"; Gosling et al., 2003), and openness to experience (e.g., "I see myself as open to new experiences, complex"; Gosling et al., 2003) on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*). These variables regarding job characteristics and personality traits were included because prior studies suggest they may influence creative performance (Baer & Oldham, 2006; Shin & Grant, 2021), especially in the context of human–AI collaboration (Tang et al., 2022).

Results

Table 3 presents the descriptive statistics and zero-order correlations between variables of interest in Study 3. To test the hypothesis, I performed hierarchical regression analyses (Aiken & West, 1991; Table 4). Following my preregistered plan, I first entered control variables including intrinsic motivation, creative requirement, conscientiousness, and openness to experience (for additional controls as well as ancillary results, see Supplemental Table S2). In the next step, I included human–AI collaboration (self-report at Time 1) to predict creativity (supervisor-rated at Time 2), finding a positive significant effect ($b = .13$, $SE = .03$, 95% CI [.06, .19], $p < .001$). In the full model, the quadratic term of human–AI collaboration was a significant predictor of creativity ($b = -.04$, $SE = .02$, 95% CI [−.07, −.004], $p = .030$). These results of both significant linear and quadratic terms indicated a nonsymmetrical inverted-U-shaped relationship (Aiken & West, 1991). As displayed in Figure 6, from a low (value of 1) to moderate (around 4 or 5) level of AI usage, increased collaboration with AI elevated creative work performance. However, according to the regression equation, a tipping point was expected to occur at a value of 5 in human–AI collaboration, with the maximum value of creativity reaching 4.5. After this point, creativity should decline. For probing robustness, I also replicated this Goldilocks effect utilizing another direct indicator of creativity for the employee's *creative work* (instead of focusing on the individual) evaluated by the supervisor. Please refer to Supplemental Tables S3–S5.

For transparency, I note that the quadratic effect of human–AI collaboration on creativity was not consistently significant when control variables were excluded from the models ($b_{\text{creativity as outcome}} = -.03$, $SE = .02$, 95% CI [−.08, .01], $p = .14$; $b_{\text{creativity of work as outcome}} = -.08$, $SE = .04$, 95% CI [−.16, −.003], $p = .043$), while the linear term remained significant. This suggests that individual differences may have suppressed or masked the curvilinear relationship in the uncontrolled models. Including theoretically relevant control variables helps to isolate the unique contribution of the quadratic term, revealing a more nuanced, nonlinear pattern that may otherwise be obscured by individual variability. In accordance with the preregistered plan to account for potential confounds, I consider the controlled models to offer a more precise representation of the relationship.

Discussion and Constraints on Generality

Established findings in the literature have predominantly centered on the positive impacts of AI on productivity (Brynjolfsson et al., 2025; Frank et al., 2019; Noy & Zhang, 2023; Wang et al., 2024), and early insights suggest a similar linear effect on creativity (Boussioux et al., 2024; Hitsuwari et al., 2023; Jia et al., 2024; Lee & Chung, 2024). What remains unclear, however, is whether a potential curvilinear effect exists—an area that has hitherto received little consideration from past scholars. This "too much of a good thing" phenomenon is not uncommon in our lived experience (Grant & Schwartz, 2011), particularly with respect to performance. For example, the relationship between extraversion and work performance in sales has been shown to follow an inverted U-shape (Grant, 2013); similarly, several factors (e.g., time pressure, procrastination) demonstrate nonlinear dynamics with employee creative performance (Baer & Oldham, 2006; Shin & Grant, 2021). As scholars caution against the danger of assuming monotonic positive or negative relationships, which at times may not realistically reflect everyday phenomena, and call for more research (Pierce & Aguinis, 2013), the present investigation fills this void and tests for a potential curvilinear effect in the context of interest.

Across three studies (two preregistered), I examine the impact of AI on human creativity using a meticulously crafted experimental paradigm with ChatGPT among business professionals in Canada and working adults in the United States, along with a multiwave, multisource field survey among American workers in the creative

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Table 3
Means, Standard Deviations, and Correlations in Study 3

Variable	*M*	*SD*	1	2	3	4	5	6
Independent variable								
1. Human–AI collaboration	2.83	1.77	—					
Dependent variable								
2. Creativity (supervisor-rated)	4.17	0.98	.36***	—				
Control variables								
3. Intrinsic motivation for job	5.31	1.35	.33***	.44***	—			
4. Creative requirement for job	4.69	1.60	.15*	.54***	.18*	—		
5. Conscientiousness personality	5.83	1.05	−.02	.06	.18*	−.02	—	
6. Openness to experience personality	5.31	1.19	−.07	.31***	.21**	.13	.22**	—

Note. $N = 188$. AI = artificial intelligence.
* $p < .05$. ** $p < .01$. *** $p < .001$.

industries. I theorize and discover an inverted-U-shaped relationship between human–AI collaboration and creative outcomes. The current research finds that both excessive and insufficient interactive collaborations with AI hinder creativity, whereas a moderate level of collaboration leads to optimal creativity. While few results (e.g., AI-rated creativity in Study 1) show somewhat similar levels of creative outcomes between moderate and high human–AI collaboration treatment, this does suggest that increased interaction with AI leads to diminishing returns without necessarily yielding greater creative benefits. At the very least, this cautiously supports my proposition of a nonmonotonic curvilinear (though not a perfectly inverted U) relationship with a modest effect size, relative to a monotonic linear effect.

Interestingly, this observed effect is only explained by knowledge diversity in lieu of problem restructuring. Findings of mediation effects imply that working with AI to a greater or lesser extent for problem-solving does not fundamentally change the way humans think or approach issues from a novel perspective (i.e., thinking out of the box). However, effective collaboration with AI in an optimal amount contributes to generating a wide array of diverse ideas containing different new knowledge, which is essential for the creative decision-making process. This step enables individuals to select the highest quality creative ideas from a variety of options and further refine them in the final stages of problem-solving (Girotra et al., 2010). My discovery of the underlying theoretical mechanism (as well as the investigation of alternative mechanisms) advances the creativity and AI literature, suggesting that an appropriate degree of collaboration with AI has the potential to influence creative performance through the production-focused creative process by "actively push[ing] ideas forward through trial and error" (i.e., idea generation behavior for better knowledge diversity) but not through the preparation-focused process by "seed[ing] the ground for breakthrough ideas to come" (i.e., problem identification behavior for better problem restructuring; Lucas & Mai, 2022, p. 3).

The current research also identifies other intriguing consequences of the Goldilocks phenomenon (see the Supplemental Material). Specifically, moderate use of AI in work tasks also makes individuals feel competent in their creative abilities. Furthermore, it fosters a sense of joint ownership of ideas between humans and AI, as if both parties are part of a team to which the idea belongs. These findings hold promise for future directions, as these psychological experiences have been found to enhance performance in organizations, including creativity and innovation (Gray et al., 2020; Tierney & Farmer, 2002). As an initial exploration, I also provide preliminary evidence that moderate human–AI collaboration results in higher felt competence and collective psychological ownership, both of which in turn increase creative performance.

In addition to the theoretical significance of advancing and reshaping our foundational knowledge of the assumed linear dynamic between human–AI collaboration and creative performance, this

Table 4
Hierarchical Regression in Study 3

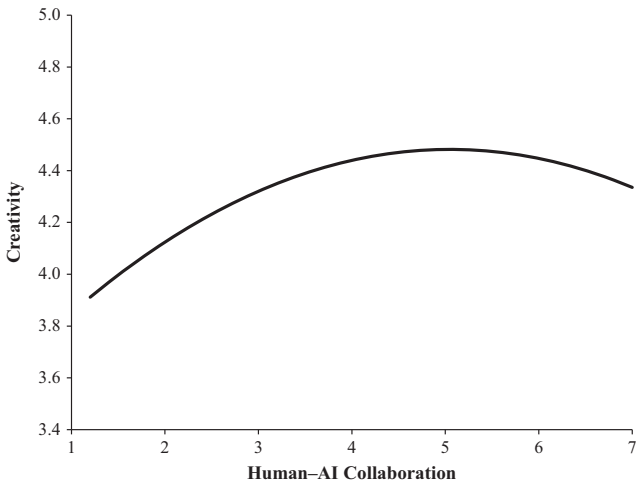
	Dependent variable: Creativity								
	Step 1: Control variable			Step 2: Adding human–AI collaboration			Step 3: Curvilinear effect of human–AI collaboration		
Predictor	*b* (*SE*)	β	*t*(183)	*b* (*SE*)	β	*t*(182)	*b* (*SE*)	β	*t*(181)
Intrinsic motivation	.23 (.04)	.32	5.56***	.17 (.04)	.24	4.10***	.18 (.04)	.25	4.33***
Creative requirement	.28 (.03)	.46	8.19***	.27 (.03)	.44	8.01***	.26 (.03)	.43	8.03***
Conscientiousness	−.02 (.05)	−.02	−0.42	−.01 (.05)	−.01	−0.25	.001 (.05)	.001	0.02
Openness to experience	.15 (.05)	.19	3.26**	.18 (.05)	.22	3.98***	.18 (.05)	.22	3.90***
Human–AI collaboration				.13 (.03)	.23	4.04***	.39 (.12)	.70	3.15**
Human–AI collaboration squared							−.04 (.02)	−.49	−2.19*
Intercept	.94 (.39)		2.38*	.75 (.38)		1.97	.34 (.42)		0.82

Note. $N = 188$. *b* = unstandardized coefficients (standard error in parentheses); β = standardized coefficients; *t* = *t*-statistic (*df*); AI = artificial intelligence; *SE* = standard error.
* $p < .05$. ** $p < .01$. *** $p < .001$.

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Figure 6

The Curvilinear (Goldilocks) Relationship Between Human–AI Collaboration and Creativity in Study 3



Note. Results are based on the hierarchical regression in Study 3 (with control variables set at their mean values). AI = artificial intelligence.

research holds some practical insights for managers and organizations. Given the importance of AI adoption for coinvention and performance growth, managers and organizations could consider not only expanding access to these technologies but also encouraging employees to engage with them in a balanced and mindful way—neither too much nor too little. As the observed findings point toward the benefits of moderate collaboration with AI, workers who collaborate effectively with AI in their daily tasks may improve creative performance at the individual level first. I believe that this effect might also manifest collectively at the organizational level over time. Interested scholars as well as practitioners could expand the present research to explore how this process unfolds from individual contributions to broader organizational outcomes.

Furthermore, this research is timely for the general public, including consumers and employees, because ChatGPT is now a widely used AI chatbot that has become integrated into daily and organizational life for numerous people in society (Berg et al., 2023; Noy & Zhang, 2023; Owens, 2023). I acknowledge that AI tools like ChatGPT may have undeniably improved our decision making and several aspects of our lives. However, these technologies also carry the potential to become a double-edged sword when excessively relied upon, as suggested by my findings alongside other studies (e.g., Chen & Chan, 2024). This further raises an interesting tension: Does overreliance on AI constrain rather than facilitate the new expression of creativity, limiting the very innovation these tools are meant to support (see the current findings on knowledge diversity and also Doshi & Hauser, 2024)? Policymakers are encouraged to pursue these broader questions and expand empirical testing at a larger scale and across different types of creative tasks. This could help form evidence-based recommendations to guide various stakeholders such as consumers, users, and workers on how to exercise caution when leveraging AI technologies, especially within the realm of creativity, to achieve the most optimal outcomes.

Some limitations of this research should be noted. First, I test the Goldilocks phenomenon across creativity assessed from a general perspective by trained judges (Studies 1 and 2), the crowd (Study 2), and AI (Studies 1 and 2); the business aspects of creative outputs evaluated by entrepreneurship professionals (Study 1); and (supervisor-rated) creative work performance amid professionals in the creative industries (Study 3). Although they mostly point to converging findings, the significant yet relatively modest correlations among them raise concerns about whether they truly capture the same underlying construct of creativity.

Prior studies offer insights into this issue, showing that experts versus crowds (Mollick & Nanda, 2016) and humans versus LLMs (Laverghetta et al., 2025) appear similar but simultaneously reveal some heterogeneity. Thus, each indicator might reflect a nuanced dimension or perspective of creativity. For example, expert judges (e.g., entrepreneurs, investors) evaluating entrepreneurial business value may emphasize profitability when assessing the quality of creative ideas (Girotra et al., 2010). Trained individuals rating general creativity may focus on the balance between originality and usefulness without taking much consideration of monetary aspects. Crowdsourced public may place weight on language features (e.g., writing quality, readability) when evaluating creativity (Mollick & Nanda, 2016). Regarding potential differences between AI and human raters, preliminary analyses using separate novelty and utility scores suggest that LLMs tend to (semantically) prioritize novelty in generating creativity composite scores, assigning greater weight to novelty than to utility, in line with some early research insights (Huang et al., 2025; Laverghetta et al., 2025). In contrast, human judges appear to use a more integrated approach overall by averaging novelty and utility equally when rating creativity. Taken together, I see these as complementary approaches to creativity in terms of robustness, since it is inherently challenging to measure creativity. At the same time, more psychological studies are needed to systematically examine how AI and human raters—particularly different types of human raters, such as experts and laypeople—evaluate creative outputs. These (mis)alignments can offer valuable insights for future researchers in selecting creativity measures that best capture the specific psychology they aim to study.

Second, my samples were mostly drawn from Western, educated, industrialized, rich, and democratic societies. As AI scholars are advocating for a more inclusive approach to champion diverse data (Mboa Nkoudou, 2023), whether this reflects universal (creative) experiences across different cultural populations should be vetted to generalize the Goldilocks effect in the real world. In addition, the mediation analyses in Studies 1 and 2 rest on the assumption that the cognitive processes during brainstorming (knowledge diversity and problem restructuring) causally affect creativity. Though temporally, brainstorming as the first stage occurs right before writing the final proposal as the second stage, readers should interpret these findings with reservation, as my studies did not manipulate the mediators and therefore could not make definite causal inferences. Future scholars could not only capitalize on moderation-of-process designs (Spencer et al., 2005) but also follow state-of-the-art recommendations for ruling out confounding through additional variables or longitudinal designs that more precisely account for temporal order within nested data to address reverse causality (Rohrer et al., 2022).

Another future direction is to explore boundary conditions or moderating factors not tested in this research (e.g., individual differences, Dygert & Jarosz, 2020; van Knippenberg & Hirst, 2020, such as skill levels, which have shown conflicting findings in AI studies; Brynjolfsson et al., 2025; Jia et al., 2024; Noy & Zhang, 2023;

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Wang et al., 2024) that may strengthen or weaken the effect. For this effort, interested researchers could refer to recent tutorials on determining satisfactory sample sizes for interaction effects (Baranger et al., 2023), rather than focusing primarily on a priori power analyses for direct effects as in the present set of studies. This is especially important given the concerning replicability rate of 37% for reported moderation in applied psychology due to insufficient statistical power (Crede & Sotola, 2024).

In summary, this research uncovers a Goldilocks effect of human–AI collaboration by documenting a curvilinear relationship with creativity in the lab and the field. I highlight the importance of striking the right balance—not too little, not too much—in AI usage to harness and unlock human creative potential. Doing so enables the generation of diverse ideas and new knowledge from both human and machine perspectives, ultimately promoting optimal creative outcomes in everyday tasks.

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